Basic Info

- **Yang Rong** · Male
  - Senior Blockchain Protocol Engineer
- **Job Preferences:** Blockchain / Web3 Industry · Remote · Full-time or Part-time
- GitHub · Notes · yrong1997@gmail.com

Professional Summary

Senior protocol engineer with 5+ years of production blockchain work in Rust, Solidity, Go, and TypeScript.

Core developer since 2021 of Snowbridge, a trustless Ethereum ↔ Polkadot bridge secured by on-chain light clients on both sides. Now building speculative messaging between parachains in Polkadot-SDK.

Earlier DeFi work at Parallel Finance: a money market and AMM on Substrate (Rust), and an NFT lending protocol on Ethereum (Solidity).

Technical Skills

- **Languages:** Rust, Solidity, Go, TypeScript
- **Blockchain:** Polkadot-SDK / Substrate (FRAME, XCM) · Ethereum and L2s (Arbitrum, Base) · upgradeable contracts · Foundry, Hardhat
- **Consensus & Cryptography:** light clients (Ethereum sync committee, BEEFY) · SSZ, Merkle and MMR proofs · Fiat-Shamir sampling
- **Distributed Systems:** relayers · idempotent delivery · finality-gated processing · indexing and reconciliation
- **DeFi:** lending, liquidation, AMMs, oracles, liquid staking
- **Tooling:** PostgreSQL, GraphQL, Docker, Nix, GitHub Actions

Working Experience

- **Snowbridge (Snowfork)** | Core Developer · 2021 – Present — *Trustless bridge between Ethereum (and L2s) and Polkadot*
  - **How it works:** No multisig or external validators. Each chain verifies the other's consensus with an on-chain light client, so the bridge is as secure as the two chains themselves.
  - **Light clients:** Develop and maintain the Ethereum beacon light client (Rust). Work on the BEEFY client (Solidity), which checks a random sample of validator signatures. It has an interactive three-transaction protocol seeded by RANDAO, and a non-interactive one-transaction Fiat-Shamir mode that I shipped.
  - **Messaging:** Work on the message queues (Rust) and the Gateway V2 upgrade (Solidity), which adds arbitrary contract calls and L2 routes.
  - **Off-chain services:** Work on the relayers (Go), including the fisherman that reports equivocating validators, and maintain the indexer and monitoring (TypeScript).

    - Snowbridge PRs · Polkadot-SDK PRs (Snowbridge pallets run upstream in the Bridge Hub runtime)

- **Polkadot-SDK: Speculative Messaging** | Contributor · 2026 – Present — *Messaging between parachains (ongoing)*

    - **What it is:** The planned successor to HRMP. Senders commit message roots per block, and the relay chain only checks that consumed roots are recent.
    - **My part:** Building the runtime and node sides as one PR stack, from relay-chain checks to end-to-end tests. Primitives are in upstream review (#12676).

- **Parallel Finance** | Blockchain Engineer · 2022 — *DeFi on Polkadot and Ethereum*

    - **Parallel (Rust / Substrate):** A Polkadot parachain with a money market, an AMM, and liquid staking in one runtime. Worked on liquidation in the money market and on the price oracles for LP and staking tokens used as collateral. Integrated EVM support and the Axelar bridge.
    - **ParaSpace (Solidity, Ethereum mainnet):** An Aave V3–based market that lends against NFTs and ERC-20s. Worked on NFT liquidation and the NFT floor-price oracle.
    - Parallel PRs · ParaSpace PRs

- **Bifrost Finance** | Blockchain Developer — *Liquid staking on Polkadot (Rust / Substrate)*

    - Worked on the SALP pallet, which gives users a tradable token for DOT locked in parachain crowdloans, and on its contribution, refund, and reward logic.

- **Subscan** | Backend Developer — *Block explorer for Substrate chains (Go)*

    - Worked on the services that index blocks and events into SQL and serve the explorer APIs. Contributed to the open-source subscan-essentials.

Education

- **USTC (University of Science and Technology of China)**
    - M.S. in Information Science & Technology
    - B.S. in Electronic Information Science & Technology

Languages

- English: CET-6. 5+ years working in English-speaking teams and international open-source projects.
- Chinese: native.